

Q1 When to retrain the data?

We retrain a model when the data distribution, target relationship, or business environment has changed enough that the current model no longer performs reliably. I would not retrain only because time has passed; I would retrain based on monitoring signals.

1. Model performance dropped.
 - a. Could due to all of the following things, also could be engineering issue.
2. Feature distribution drift
 - a. checking methods: PSI, KL divergence, KS test, mean/std shift, missing rate changes, category distribution changes
3. Label distribution drift
 - a. distribution, similar to feature
4. Concept drift (x&y)
 - a. relationship between x& y
5. business change

Best practices: retrain on monthly by default, but trigger earlier retraining if performance drop or drift.

Population Stability Index (PSI)

Feature → bins → % check for each bin → PSI

$$PSI = \sum (Actual\% - Expected\%) \times \ln\left(\frac{Actual\%}{Expected\%}\right)$$

<https://medium.com/model-monitoring-psi/population-stability-index-psi-ab133b0a5d42>

KL Divergence

Asymmetric, PSI is symmetric.

$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$

$$D_{KL}(P_{current}||P_{train}) = \sum_i P_{current}(i) \log \frac{P_{current}(i)}{P_{train}(i)}$$

Zero denominator issue: laplace smoothing. add a small fraction. or when count always add 1.

Bin num could be 20, normally 10~30

Joint distribution shift

Most practical: build a classifier to distinguish training v.s. inference AUC>0.8, shift.

Others: kernel distance measurement.

Others

What if we have online learning (new batch training everyday?).

- current inference v.s. initialized pretrain
- current inference v.s. recent batch training
- current inference v.s. recent production inference.

Would retrain on a certain frequency be enough? - Not enough

- sudden change might not be captured: a new feature launch, pipeline issue.
- Best practice is **fixed retrain + detection based retrain.**

What if actual labels are delayed? Like LT90.

- Find proper proxy e.g. LT14. Exam if LT14 is a good one for LT90.